

Spanning Tree Protocol (Toolkit)

- Topics covered:
 - PortFast
 - BPDU Guard
 - BPDU Filter

PortFast (STP)

◆ What is PortFast?

PortFast allows switch ports connected to end hosts to immediately enter the STP Forwarding state, bypassing Listening and Learning.

This lets hosts start communicating over the network as soon as they connect – no need to wait.

Cisco expects you to know other STP features:

- BPDU Guard
- BPDU Filter
- Root Guard
- Loop Guard

These will be covered later.

This video focuses on PortFast.

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◆ The problem PortFast solves

PC connected to a switch (default STP behaviour):



When an end

host connects to a switch port like this:

- The port becomes up/up
- But it cannot send or receive data yet

In SHOW IP INTERFACE BRIEF:

- G0/1 is up/up
- Frames from the PC are discarded

Why?

- The port is an STP designated port
- It takes 30 seconds to enter forwarding

Breakdown:

- 15 seconds → Listening
- 15 seconds → Learning

Listening (15s) → Learning (15s) → Forwarding

This causes poor user experience.

- The user expects the network to work immediately
- The user doesn't know STP exists
- They just think "the internet doesn't work"

This wait is unnecessary because:

- There is no Layer 2 loop risk between a switch and a PC
- PCs don't flood frames like switches

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Real switch behaviour without PortFast:

After connecting the PC cable:

- Link light turns amber
- Port is up and receiving data
- STP is blocking the data

After 30 seconds:

- Light turns green
- Port enters Forwarding
- Data can flow

◆ How PortFast solves the problem

With PortFast configured:

- The port immediately enters Forwarding
- Listening and Learning are bypassed
- No 30-second wait



→ Forwarding immediately

This provides a much better user experience.

◆ Configuring PortFast – Method 1 (Interface mode)

CLI (image shown once)

```
SW1(config)# interface g0/1
```

```
SW1(config-if)# spanning-tree portfast
```

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```
SW1(config)# interface g0/1
SW1(config-if)# spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
```

PortFast should NOT be configured on ports connected to switches or temporary Layer 2 loops can occur.

Even if you configure **spanning-tree portfast** on a trunk port, it won't be active.

Warning message shown:

- PortFast should only be enabled on ports connected to a single host
- Enabling it on switches can cause temporary bridging loops
- Use with caution

Important points:

- Do not enable PortFast on ports connected to switches
- Listening and learning exist to prevent loops
- PortFast bypasses these states

Another message:

- PortFast only works in non-trunking mode
- If configured on a trunk, it won't be active

Verifying PortFast on an interface CLI (image shown once)

“SHOW SPANNING-TREE INTERFACE G0/1 DETAIL”

```
SW1# show spanning-tree interface g0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 32769, address 5254.0016.c410
Designated bridge has priority 32769, address 5254.0016.c410
Designated port id is 128.3, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast edge mode
Link type is point-to-point by default
BPDU: sent 1272, received 0
```

show spanning-tree interface interface-name detail

- There are two kinds of PortFast:
 - edge
 - network
- edge is the kind we are covering in this video.
- network is used for a feature called Bridge Assurance (not a CCNA topic).

Output shows:

- “The port is in the PortFast edge mode”

Explanation:

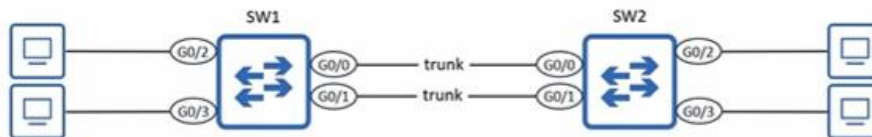
- Two kinds of PortFast:
 - Edge

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- Network
 - This video covers PortFast edge
 - PortFast network is used for Bridge Assurance (not CCNA)

◆ Configuring PortFast – Method 2 (Global mode)

Example topology



- G0/0 and G0/1 → trunks
- G0/2 and G0/3 → access ports

CLI

“SW1(config)# spanning-tree portfast default”

```
SW1(config)# spanning-tree portfast default
```

```
%Warning: this command enables portfast by default on all interfaces. You
should now disable portfast explicitly on switched ports leading to hubs,
switches and bridges as they may create temporary bridging loops.
```

- Access ports only (not trunk ports).
- To disable PortFast on a specific access port:
 - SW1(config-if)# spanning-tree portfast disable

Warning message:

- Enables PortFast by default on all interfaces
- Disable PortFast on ports connected to switches

Clarification:

- PortFast is only activated on access ports
- Not activated on trunk ports
- Warning text is misleading

To disable on a specific access port:

“SW1(config-if)# spanning-tree portfast disable”

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Verifying global PortFast behaviour

Access port (G0/2)

“SHOW SPANNING-TREE INTERFACE G0/2 DETAIL”

```
SW1# show spanning-tree interface g0/2 detail
Port 3 (GigabitEthernet0/2) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.3.
Designated root has priority 32769, address 5254.0016.c410
Designated bridge has priority 32769, address 5254.0016.c410
Designated port id is 128.3, designated path cost 0
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 4
The port is in the portfast edge mode by default
Link type is point-to-point by default
BPDU: sent 22, received 0
```

Output:

- “The port is in the portfast edge mode by default”

Trunk port (G0/1)

“SHOW SPANNING-TREE INTERFACE G0/1 DETAIL”

```
SW1# show spanning-tree interface g0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0001 is designated blocking
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 32769, address 5254.0016.c410
Designated bridge has priority 32769, address 5254.0016.c410
Designated port id is 128.2, designated path cost 0
Timers: message age 0, forward delay 10, hold 0
Number of transitions to forwarding state: 0
Link type is point-to-point by default
BPDU: sent 17, received 0
```

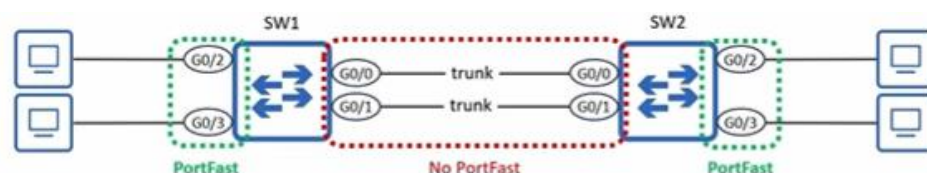
Output:

- No mention of PortFast

Key point:

- Global PortFast affects access ports only

Final Topology:



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◆ PortFast on trunk ports

Standard commands:

- Enable PortFast only on access ports

Some valid trunk use cases:

1. Virtualization server with VMs in multiple VLANs
2. Router-on-a-stick



Routers:

- Do not flood frames
- Do not cause Layer 2 loops

Configuring PortFast on a trunk

CLI (image shown once)

```
SW1(config)# interface g0/1
```

```
SW1(config-if)# spanning-tree portfast trunk
```

```
SW1(config-if)# spanning-tree portfast trunk
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

SW1# show spanning-tree interface g0/1 detail
!output omitted
The port is in the portfast edge trunk mode
!output omitted
```

Verification:

SHOW SPANNING-TREE INTERFACE G0/1 DETAIL

Output:

- “The port is in portfast edge trunk mode”

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◆ PortFast EDGE keyword behavior

Two modes:

- PortFast edge (CCNA)
- PortFast network (not CCNA)

Automatic behavior:

- EDGE is added automatically

Examples:

- spanning-tree portfast
→ spanning-tree portfast edge
- spanning-tree portfast trunk
→ spanning-tree portfast edge trunk
- spanning-tree portfast default
→ spanning-tree portfast edge default

To configure network mode:

spanning-tree portfast network

Exception:

- spanning-tree portfast disable does not use EDGE

Verifying EDGE in running-config

CLI (image shown once)

“SHOW RUNNING-CONFIG INTERFACE G0/1”

```
SW1(config)# interface g0/1
SW1(config-if)# spanning-tree portfast

SW1# show running-config interface g0/1
Building configuration...

Current configuration : 113 bytes
!
interface GigabitEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast edge
end
```

Use show running-config interface interface-name to view the running-config for the specified interface.
Doesn't work in Packet Tracer.

Automatically added edge to the end of the command.

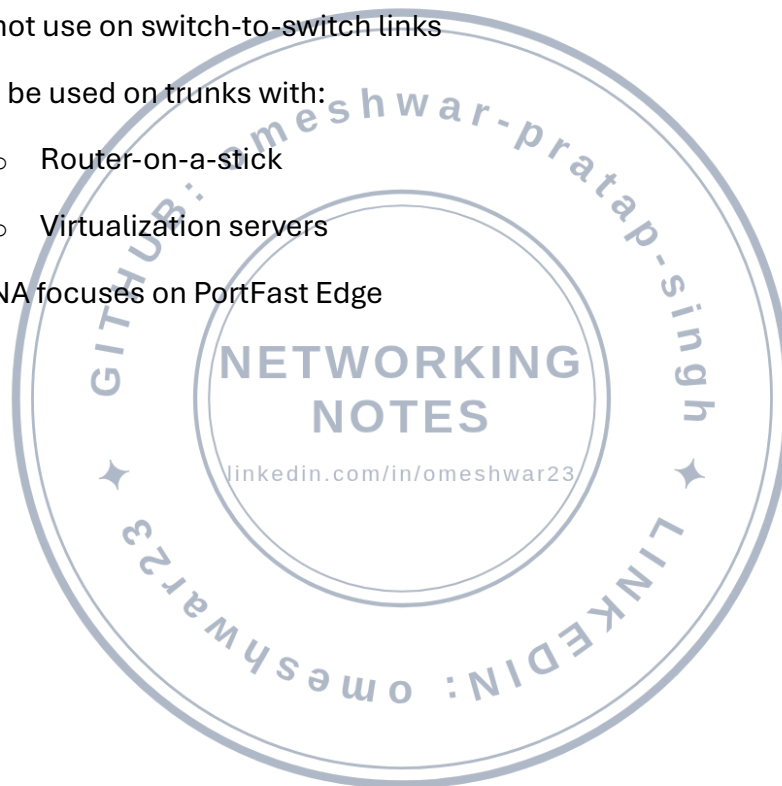
Result:

- EDGE keyword added automatically
- Command does not work in Packet Tracer

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◆ Summary

- By default, a port takes 30 seconds to forward traffic
- PortFast allows immediate forwarding
- Improves user experience
- Two configuration methods:
 - Interface mode: spanning-tree portfast
 - Global mode: spanning-tree portfast default
- Only active on access ports
- Do not use on switch-to-switch links
- Can be used on trunks with:
 - Router-on-a-stick
 - Virtualization servers
- CCNA focuses on PortFast Edge



BPDUGuard

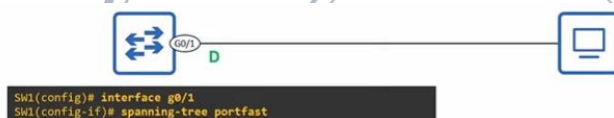
◆ Introduction

In this, we cover two more STP features:

- **BPDUGuard**
Automatically disables a switch port if it receives a BPDU.
- **BPDUGuard Filter**
Stops a port from sending BPDUs or processing received BPDUs.

◆ How a PortFast-enabled port handles BPDUs

PortFast makes a port start in the **Forwarding** state when it is connected, but **it does not disable STP** on the port.



So:

- The port still **sends BPDUs**
- It sends them **every 2 seconds**, like a normal STP port

PortFast port connected to a PC



- The PC does not run STP
- The PC ignores the BPDUs
- The switch keeps sending them

Verification (CLI image)

“SHOW SPANNING-TREE INTERFACE G0/1 DETAIL”

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```
SW1(config-if)# do show spanning-tree interface g0/1 detail
!output omitted
The port is in the portfast edge mode
Link type is point-to-point by default
BPDU: sent 77, received 0
```

Observed:

- PortFast is enabled on G0/1
- BPDU sent: 77
- BPDU received: 0

Because end hosts don't run STP:

- A PortFast-enabled port should not receive BPDUs

◆ What if a PortFast port receives a BPDU?

PortFast port connected to another switch



- When a switch comes online, it assumes it is the root bridge
- It sends its own BPDUs

If a PortFast-enabled port receives a BPDU:

- It reverts to acting like a regular STP port
- PortFast is no longer effective

Because:

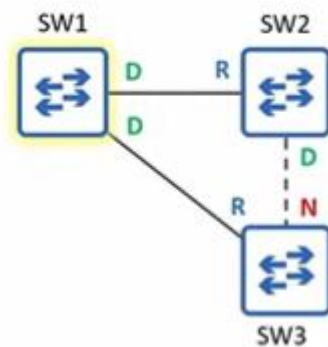
- SW2 has a lower priority
- SW2 becomes the root bridge
- SW1 G0/1 becomes a root port



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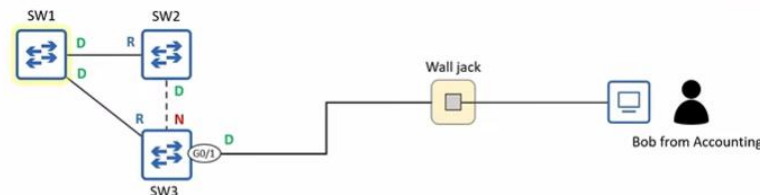
Why this is a problem (real LAN example)

Original STP topology



- SW1 is the root bridge
- SW2 and SW3 have root ports toward SW1
- Link between SW2 and SW3 is blocked

Wall jack and user access

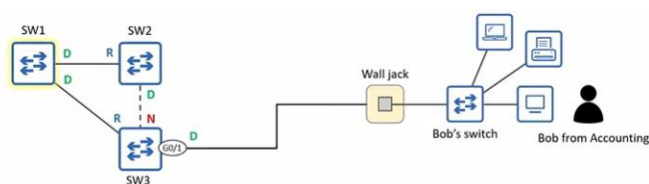


Users don't connect directly to switches.
They connect through a **wall jack**.

Bob's mistake

Bob:

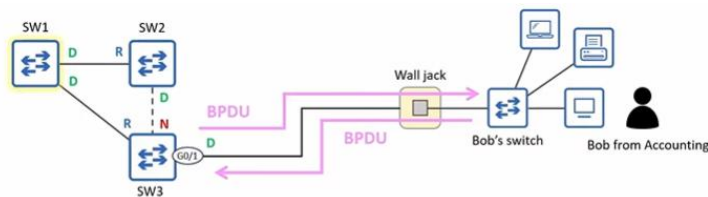
- Brings his own switch from home
- Connects it to the wall jack
- Connects laptop and printer to his switch



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Now:

- Bob's switch and SW3 exchange BPDUs



- This can affect the STP topology

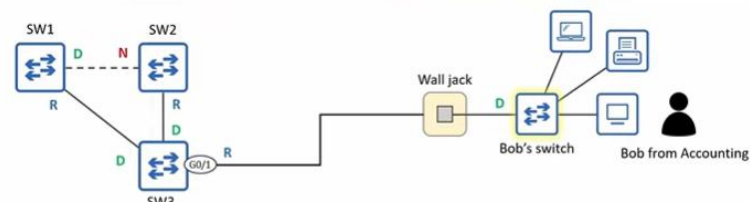
Worst-case scenario

If Bob's switch:

- Has a lower bridge ID
- Becomes the root bridge

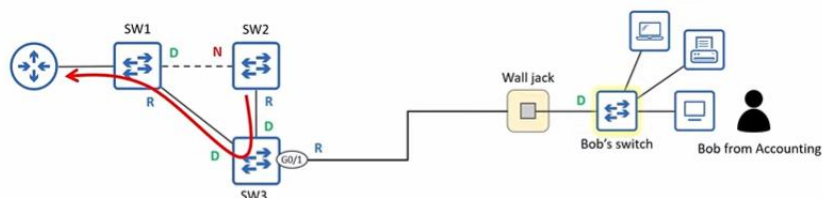
Then:

- All switches point their root ports toward Bob's switch
- SW1–SW2 link becomes blocked



Result:

- Traffic takes a less efficient path
- Unauthorized device affects the network



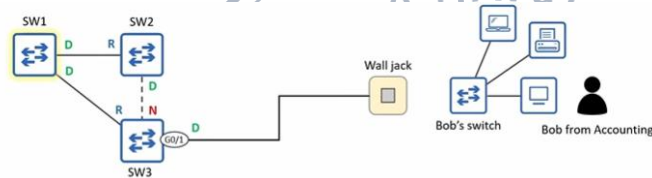
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Summary of the problem

- PortFast should only be enabled on ports connected to non-switch devices
- End hosts do not send BPDUs
- A PortFast port still sends BPDUs
- If it receives BPDUs, it behaves like a normal STP port
- A user connecting a switch can affect STP

Shame on you, Bob.

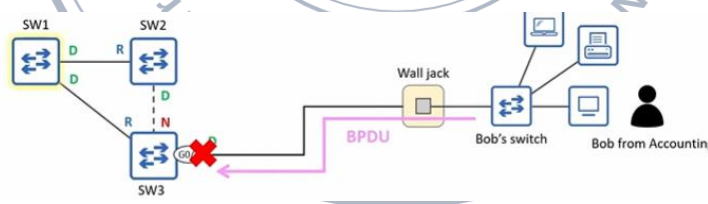
◆ BPDU Guard



BPDU Guard protects the network from unauthorized switches.

What BPDU Guard does

- The port continues sending BPDUs
- If it **receives a BPDU**, it enters the **err-disabled state**



Effect:

- The port is disabled
- No data or BPDUs can be sent or received
- The unauthorized switch cannot affect STP

Bob's result:

- “The internet doesn’t work”
- Bob complains to IT

Mission accomplished.

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◆ Configuring BPDU Guard (per-port)

CLI (image shown once)

```
SW3(config)# interface g0/1
SW3(config-if)# spanning-tree bpduguard enable
SW3(config-if)# do show spanning-tree interface g0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 24577, address 5254.001a.9d29
Designated bridge has priority 32769, address 5254.0006.448f
Designated port id is 128.2, designated path cost 4
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast edge mode
Link type is point-to-point by default
Bpdu guard is enabled
BPDU: sent 93, received 0
```

“interface g0/1”

“spanning-tree bpduguard enable”

Verification:

SHOW SPANNING-TREE INTERFACE G0/1 DETAIL

Observed:

- BPDU Guard is enabled

◆ Configuring BPDU Guard (default)

CLI

“spanning-tree portfast bpduguard default”

```
SW3(config)# spanning-tree portfast bpduguard default
SW3(config)# do show spanning-tree interface g0/1 detail
Port 2 (GigabitEthernet0/1) of VLAN0001 is designated forwarding
Port path cost 4, Port priority 128, Port Identifier 128.2.
Designated root has priority 24577, address 5254.001a.9d29
Designated bridge has priority 32769, address 5254.0006.448f
Designated port id is 128.2, designated path cost 4
Timers: message age 0, forward delay 0, hold 0
Number of transitions to forwarding state: 1
The port is in the portfast edge mode
Link type is point-to-point by default
Bpdu guard is enabled by default
BPDU: sent 165, received 0
```

spanning-tree portfast [edge] bpduguard default

- When enabled by default, BPDU Guard is activated on all PortFast-enabled ports.
- Use **spanning-tree bpduguard disable** in interface config mode to disable it on specific ports.

Important difference:

- PortFast default → all access ports
- BPDU Guard default → all PortFast-enabled ports

Not necessarily all access ports.

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To disable on a specific port:

“spanning-tree bpduguard disable”

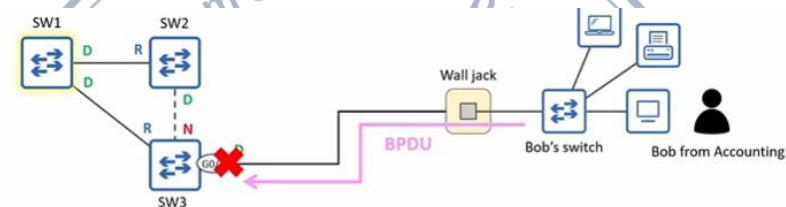
◆ BPDU Guard violation (err-disable)

When a BPDU is received:

- Port enters err-disabled state

Messages seen in CLI:

- Received BPDU with BPDU Guard enabled
- Putting interface into err-disable state
- Interface changed state to down/down



```
*Sep 3 05:08:11.977: %SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port GigabitEthernet0/1 with BPDU Guard enabled. Disabling port.
*Sep 3 05:08:11.977: %PM-4-ERR_DISABLE: bpduguard error detected on Gi0/1, putting Gi0/1 in err-disable state
*Sep 3 05:08:12.978: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
*Sep 3 05:08:13.978: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
```

Verification:

“SHOW INTERFACES G0/1”

```
SW3# show interfaces g0/1
GigabitEthernet0/1 is down, line protocol is down (err-disabled)
!output omitted
```

Observed:

- Interface is err-disabled

ErrDisable:

- Cisco feature that disables ports under certain conditions
- BPDU Guard violation is one such condition

◆ Err-disabled port behaviour

- Link light turns solid amber
- Disconnecting the cable does **not** fix it

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- The port remains disabled

◆ Re-enabling an err-disabled port

First:

- Fix the underlying issue
(disconnect Bob's switch)

Then:

Two methods:

1. Manual

- shutdown
- no shutdown

2. Automatic

- ErrDisable Recovery

◆ ErrDisable Recovery

ErrDisable Recovery:

- Automatically re-enables err-disabled ports
- After a set period of time

Verification (CLI image)

“SHOW ERDDISABLE RECOVERY”

```
SW3# show errdisable recovery
ErrDisable Reason    Timer Status
-----
arp-inspection       Disabled
bpduguard            Disabled
channel-misconfig (STP) Disabled
output omitted

Timer interval: 300 seconds

Interfaces that will be enabled at the next timeout:
```

ErrDisable Recovery is disabled by default.

- The default recovery timer is 300 seconds (5 minutes).
 - Err-disabled interfaces will be automatically re-enabled after 5 minutes.
- Use SW1(config)# **errdisable recovery interval seconds** to modify the interval.

Observed:

- BPDU Guard listed as a cause
- ErrDisable Recovery is disabled by default

Default timer: 300 seconds (5 minutes)

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◆ Enabling ErrDisable Recovery for BPDU Guard

CLI (image shown once)

“errdisable recovery cause bpduguard”

Verification:

“SHOW ERRDISABLE RECOVERY”

```
SW3(config)# errdisable recovery cause bpduguard
SW3(config)# do show errdisable recovery
ErrDisable Reason    Timer Status
-----
arp-inspection        Disabled
bpduguard              Enabled
channel-misconfig (STP) Disabled

Timer interval: 300 seconds

Interfaces that will be enabled at the next timeout:
Interface    Errdisable reason    Time left(sec)
-----
Gi0/1         bpduguard            296
```

→ G0/1 will be automatically re-enabled after 296 seconds.

Observed:

- BPDU Guard enabled
- G0/1 listed
- Time left: 296 seconds

When timer reaches 0:

- Interface is automatically re-enabled

Important:

- If Bob's switch is still connected
- The port will be err-disabled again

◆ Final point

Even with ErrDisable Recovery:

- You must fix the underlying problem
- Otherwise the port will be disabled again

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◆ Conclusion

- BPDU Guard protects the network from unauthorized switches
- It disables the port if a BPDU is received
- Commonly used with PortFast
- ErrDisable Recovery can automatically restore ports
- The problem must be fixed first

That completes **BPDU Guard**.



BPDU Filter (STP)

♦ Moving on to BPDU Filter

Now let's move on to **BPDU Filter**.

Before looking at how it works, let's see the **problem it solves**.

A switch port connected to an end host:

- Continues sending **BPDUs every 2 seconds**
- This happens **regardless of whether PortFast and BPDU Guard are enabled**

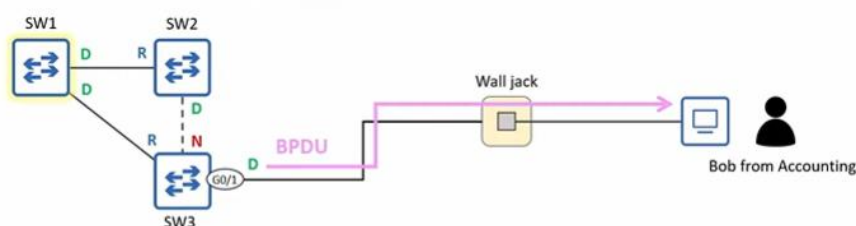
So even though Bob's PC:

- Doesn't run spanning tree
- Ignores any BPDUs

SW3:

- Keeps sending BPDUs to Bob's PC
- Once every 2 seconds

BPDU sending to an end host (problem)



If the port does not connect to a switch:

- Sending BPDUs is unnecessary
- Sending BPDUs is undesirable

Reasons:

1. BPDUs use some bandwidth and processing power (minimal, but unnecessary)

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2. BPDUs contain information about the LAN's STP topology
If maximum security is a concern, this information should not be sent to user devices

BPDU Filter solves this by preventing a port from sending BPDUs.

◆ How BPDU Filter works

BPDU Filter:

- Stops a port from sending BPDUs

If enabled on SW3 G0/1:

- It will not send BPDUs to Bob's PC anymore

Unlike BPDU Guard:

- BPDU Filter does **not** disable the port if it receives a BPDU

How the port reacts to receiving a BPDU:

- Depends on how BPDU Filter is configured

◆ Ways to configure BPDU Filter

BPDU Filter can be enabled in two ways:

1. Per-port (interface config mode)
2. By default (global config mode)

Unlike PortFast and BPDU Guard:

- BPDU Filter behaves differently depending on the method used

BPDU Filter per-port (interface config mode)

Command:

"SPANNING-TREE BPDUFILTER ENABLE"

Behaviour:

- The port does not send BPDUs
- The port ignores any BPDUs it receives

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Effect:

- STP is effectively disabled on the port

Important warning:

- Use with caution (bold red letters)
- BPDU Filter on the wrong port can cause permanent Layer 2 loops
- This can cause a broadcast storm
- This can bring down the network

STP prevents loops:

- Disabling STP on a port is very risky

Recommended method: BPDU Filter by default

A better choice:

- Enable BPDU Filter by default in global config mode

Command:

“SPANNING-TREE PORTFAST BPDUFILTER DEFAULT”

Notes:

- EDGE keyword can optionally be used
- BPDU Filter is activated on all PortFast-enabled ports
- Not on all access ports

You can disable it on specific ports if needed:

“SPANNING-TREE BPDUFILTER DISABLE”

Default BPDU Filter behavior (important difference)

When BPDU Filter is enabled **by default**:

Normal state:

- The port does **not send BPDUs**

If the port **receives a BPDU**:

- PortFast is disabled

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- BPDU Filter is disabled
- The port operates as a **normal STP port**

This is the best of both worlds:

- No BPDUs sent under normal conditions
- Proper STP behavior if BPDUs are received

Recommendation

- PortFast and BPDU Guard can be enabled per-port or by default
- Their behavior does not change based on the method

However:

- BPDU Filter should only be enabled by default
- Avoid enabling it per-port unless you have a very good reason

Disabling STP on a port:

- Is risky business

◆ BPDU Guard and BPDU Filter together

They can be enabled on the same port.

Result depends on how BPDU Filter is configured.

BPDU Filter enabled by default (global)

If the port receives a BPDU:

- BPDU Filter is disabled
- BPDU Guard is triggered
- The interface is err-disabled

BPDU Filter enabled per-port (interface)

If the port receives a BPDU:

- The BPDU is ignored
- BPDU Guard is not triggered
- BPDU Guard has no effect

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◆ One-slide summary

- PortFast should only be enabled on ports connected to non-switch devices
- These devices do not send BPDUs
- The port still sends BPDUs and behaves like a normal STP port if it receives BPDUs
- A user connecting a switch can affect the STP topology
- BPDU Guard protects against unauthorized switches
- BPDU Guard err-disables the port if a BPDU is received
- BPDU Guard can be enabled per-port or by default
- Err-disabled ports can be re-enabled manually or with ErrDisable Recovery
- BPDU Filter prevents a port from sending BPDUs
- BPDU Filter per-port ignores received BPDUs (very risky)
- BPDU Filter by default disables itself if BPDUs are received
- BPDU Filter is safest when enabled by default

